EOG Filter VEOG and HEOG R12 for S5S6 Trial 1

Setup the path and load data from the .vhdr file

Using a specific interval

```
clearvars; close all; clc;

addpath(genpath('Bali_Data_and_Codes'));
if ~exist('pop_loadbv','file'), eeglab; close; end
```

```
Disable MATLAB Copilot in settings to hide "Explain Error" buttons
Some menu items hidden. Use Preference menu to show them all.
eeglab: options file is C:\Users\jvela\eeg_options.m
Retrieving plugin versions from server...
Retrieving download statistics...
EEGLAB: adding "EEG-BIDS" v10.4 (see >> help eegplugin_eegbids) - new version 10.5 available
EEGLAB: adding "ICLabel" v1.7 (see >> help eegplugin_iclabel)
EEGLAB: adding "bva-io" v1.74 (see >> help eegplugin_bva_io) - new version 1.75 available
EEGLAB: adding "clean_rawdata" v2.11 (see >> help eegplugin_clean_rawdata)
EEGLAB: adding "dipfit" v5.6 (see >> help eegplugin_dipfit)
EEGLAB: adding "firfilt" v2.8 (see >> help eegplugin_firfilt)
EEGLAB: adding "lsl_app_matlabviewer" v1.3 (see >> help eegplugin_lsl_app_matlabviewer)
Disable MATLAB Copilot in settings to hide "Explain Error" buttons
Warning:
A newer revision of EEGLAB (v2026.0.0) is available HERE.

See Release notes for more information
You may disable this message in the File > Preferences menu.
```

```
filePath = 'C:\Users\jvela\Downloads\BGBMB - Rehearsal 12 (4 Jul, 2024)\BGBMB -
Rehearsal 12 (4 Jul, 2024)';
fileName = '20240704_R12_S5S6_EEG_Run2.vhdr';

% Trial 1
target_interval = [ 1568881 1998272];
```

Load and print the EOG channels for each subject

```
EEG_S5 = pop_loadbv(filePath, fileName, target_interval, [1:32]);
```

```
pop_loadbv(): reading header file
Done.
pop_loadbv(): reading EEG data
The header file points to a binary file that does not exist
trying to open a binary file with the same name as the header
file and a different extension...
Binary file is truncated/incomplete
pop_loadbv(): scaling EEG data
pop_loadbv(): reading marker file
Done.
pop_loadbv() warning: data files in header and marker files inconsistent.
```

```
EEG_S6 = pop_loadbv(filePath, fileName, target_interval, [33:64]);
```

```
pop_loadbv(): reading header file
Done.
pop_loadbv(): reading EEG data
```

```
The header file points to a binary file that does not exist
trying to open a binary file with the same name as the header
file and a different extension...
Binary file is truncated/incomplete
pop_loadbv(): scaling EEG data
pop_loadbv(): reading marker file
Done.
pop_loadbv() warning: data files in header and marker files inconsistent.
```

```
disp('Channels for S5:');
```

```
Channels for S5:
```

```
disp({EEG_S5.chanlocs.labels}');
```

```
    {'1' }
    {'2' }
    {'3' }
    {'4' }
    {'5' }
    {'6' }
    {'7' }
    {'8' }
    {'9' }
    {'10'}
    {'11'}
    {'12'}
    {'13'}
    {'14'}
    {'15'}
    {'16'}
    {'17'}
    {'18'}
    {'19'}
    {'20'}
    {'21'}
    {'22'}
    {'23'}
    {'24'}
    {'25'}
    {'26'}
    {'27'}
    {'28'}
    {'29'}
    {'30'}
    {'31'}
    {'32'}
```

```
disp('Channels for S6:');
```

```
Channels for S6:
```

```
disp({EEG_S6.chanlocs.labels}');
```

```
    {'33'}
    {'34'}
    {'35'}
    {'36'}
    {'37'}
    {'38'}
    {'39'}
```

```
{'40'}
{'41'}
{'42'}
{'43'}
{'44'}
{'45'}
{'46'}
{'47'}
{'48'}
{'49'}
{'50'}
{'51'}
{'52'}
{'53'}
{'54'}
{'55'}
{'56'}
{'57'}
{'58'}
{'59'}
{'60'}
{'61'}
{'62'}
{'63'}
{'64'}
```

Extract index for EOG channels

```
s5_L_idx = 31;
s5_R_idx = 32;
s6_L_idx = 31;
s6_R_idx = 32;
v_top_idx = 15;
v_bot_idx = 20;
```

Calculate the Bipolar Derivation **HEOG and VEOG**

```
heog_s5_raw = double(EEG_S5.data(s5_L_idx, :) - EEG_S5.data(s5_R_idx, :));
veog_s5_raw = double(EEG_S5.data(v_top_idx, :) - EEG_S5.data(v_bot_idx, :));

heog_s6_raw = double(EEG_S6.data(s6_L_idx, :) - EEG_S6.data(s6_R_idx, :));
veog_s6_raw = double(EEG_S6.data(v_top_idx, :) - EEG_S6.data(v_bot_idx, :));
```

Constants

```
fs = EEG_S5.srate;
N = length(heog_s5_raw);
t = (0:N-1) / fs;
```

Butterworth Filter 4th order

`butter` function generates a **Butterworth filter**. This is a recursive filter, meaning it uses its own previous output to calculate the next output (feedback)

b: feedforward

a: feedback

FIR (Finite Impulse Response) filters use only current and past input samples for output, while IIR (Infinite Impulse Response) filters also use past output samples, allowing for more complex filtering with fewer coefficients. FIR filters are generally more stable and have a linear phase response, whereas IIR filters can be more efficient but may face stability issues

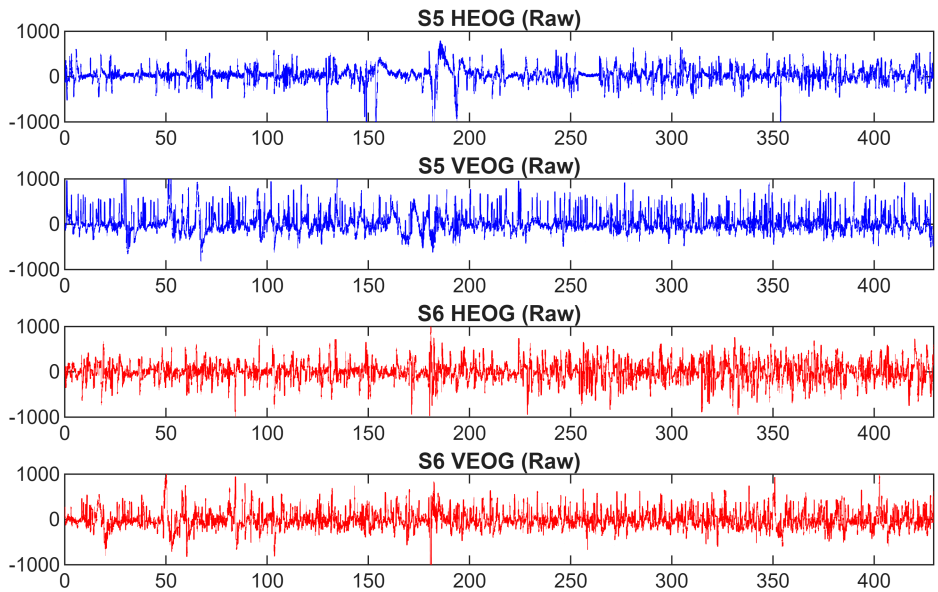
```
low_cut = 0.1;
high_cut = 30;
filter_order = 4;

[z, p, k] = butter(filter_order, [low_cut high_cut]/(fs/2), 'bandpass');
[sos, g] = zp2sos(z, p, k);
```

Zero Phase Filtering to cancel time delay

```
heog_s5_clean = filtfilt(sos, g, heog_s5_raw);
veog_s5_clean = filtfilt(sos, g, veog_s5_raw);

heog_s6_clean = filtfilt(sos, g, heog_s6_raw);
veog_s6_clean = filtfilt(sos, g, veog_s6_raw);
```

Plot raw signals for the whole time period

```
figure('Color','w', 'Name', 'Full Raw Signals'); ylabel('Amplitude (\muV)');
xlabel('Time (s)');
standard_limit = 1000;
ylim_standard = [-standard_limit, standard_limit];
subplot(4,1,1); plot(t, heog_s5_raw, 'b'); title('S5 HEOG (Raw)'); xlim([0
max(t)]); ylim(ylim_standard);
subplot(4,1,2); plot(t, veog_s5_raw, 'b'); title('S5 VEOG (Raw)'); xlim([0
max(t)]); ylim(ylim_standard);
subplot(4,1,3); plot(t, heog_s6_raw, 'r'); title('S6 HEOG (Raw)'); xlim([0
max(t)]); ylim(ylim_standard);
subplot(4,1,4); plot(t, veog_s6_raw, 'r'); title('S6 VEOG (Raw)'); xlim([0
max(t)]); ylim(ylim_standard);
```



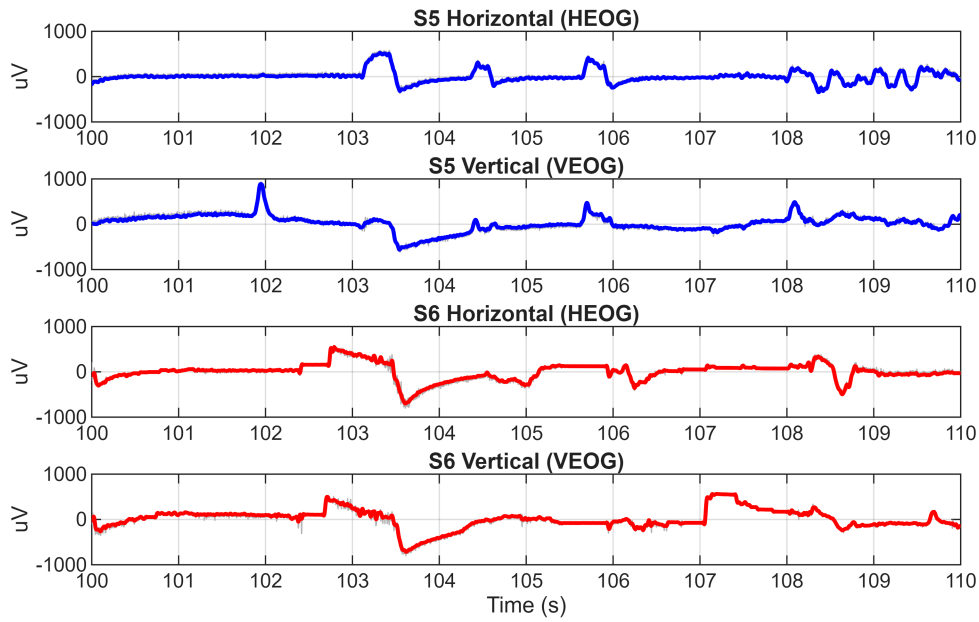
Plot filtered signals vs raw signals for comparision

Add time_axis(idx) for only zooming into a window

Leave 't' alone for the whole time

```
time_axis = t;
window_sec = 100:110; % choosen window
idx = round(window_sec(1)*fs) : round(window_sec(end)*fs);
zoom_ylim = [-1000 1000];
zoom_xlim = [100 110];
figure('Color','w', 'Name', 'Filtered Signals VEOG and HEOG');

subplot(4,1,1);
plot(time_axis(idx), heog_s5_raw(idx), 'Color', [0.7 0.7 0.7]); hold on;
plot(time_axis(idx), heog_s5_clean(idx), 'b', 'LineWidth', 1.5);
title('S5 Horizontal (HEOG)'); ylabel('uV'); grid on; xlim(zoom_xlim);
ylim(zoom_ylim);

subplot(4,1,2);
plot(time_axis(idx), veog_s5_raw(idx), 'Color', [0.7 0.7 0.7]); hold on;
plot(time_axis(idx), veog_s5_clean(idx), 'b', 'LineWidth', 1.5);
title('S5 Vertical (VEOG)'); ylabel('uV'); grid on; xlim(zoom_xlim);
ylim(zoom_ylim);

subplot(4,1,3);
plot(time_axis(idx), heog_s6_raw(idx), 'Color', [0.7 0.7 0.7]); hold on;
plot(time_axis(idx), heog_s6_clean(idx), 'r', 'LineWidth', 1.5);
title('S6 Horizontal (HEOG)'); ylabel('uV'); grid on; xlim(zoom_xlim);
ylim(zoom_ylim);
```

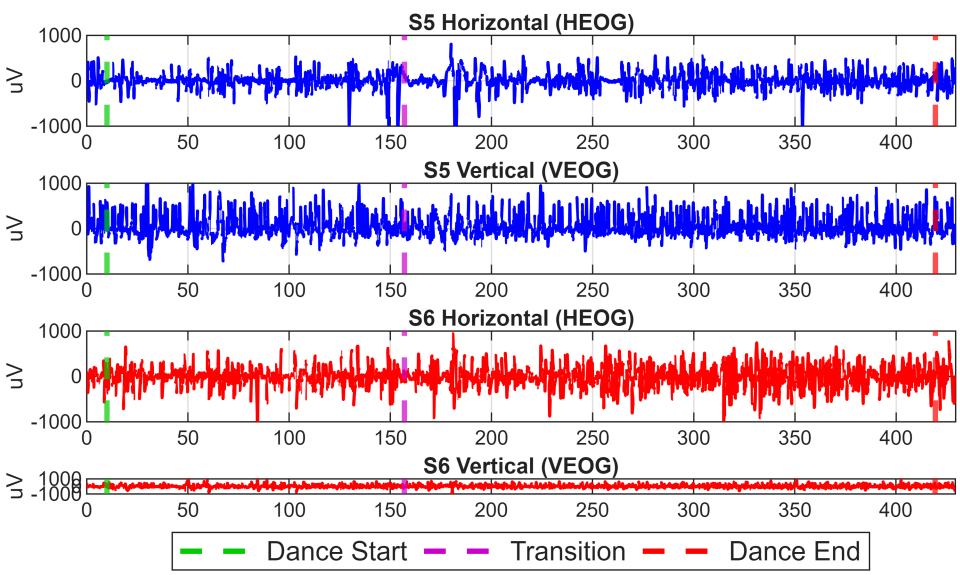
```
subplot(4,1,4);
plot(time_axis(idx), veog_s6_raw(idx), 'Color', [0.7 0.7 0.7]); hold on;
plot(time_axis(idx), veog_s6_clean(idx), 'r', 'LineWidth', 1.5);
title('S6 Vertical (VEOG)'); ylabel('uV'); grid on; xlim(zoom_xlim);
ylim(zoom_ylim); xlabel('Time (s)');
```



Marker annotation on complete clean signal for the whole time period

```
figure('Color','w', 'Name', 'Filtered Signals Only (Clean)');
zoom_ylim = [-1000 1000];

events = {
    1578881, 'Dance Start',              [0 0.8 0];   % Green
    1725881, 'Transition',               [0.8 0 0.8]; % Magenta
    1988272, 'Dance End',                [1 0 0]      % Red
};

current_interval_start = target_interval(1);

for sp = 1:4
    subplot(4,1,sp);

    if sp == 1, data_clean = heog_s5_clean; title_str = 'S5 Horizontal (HEOG)';
    elseif sp == 2, data_clean = veog_s5_clean; title_str = 'S5 Vertical (VEOG)';
    elseif sp == 3, data_clean = heog_s6_clean; title_str = 'S6 Horizontal (HEOG)';
    elseif sp == 4, data_clean = veog_s6_clean; title_str = 'S6 Vertical (VEOG)';
xlabel('Time (s)');
    end

    if sp <= 2
        line_color = 'b';
```

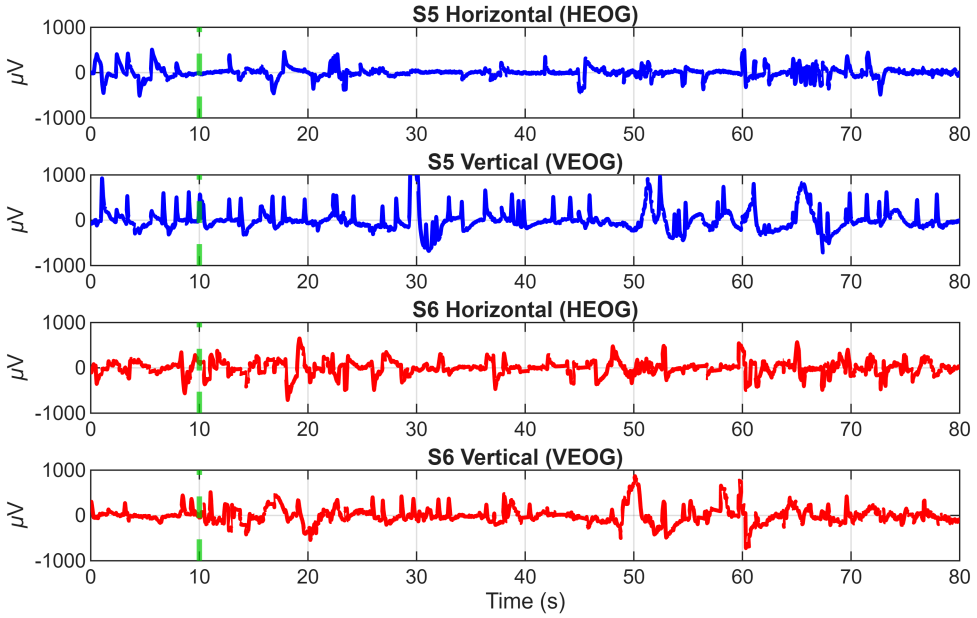
```
    else
        line_color = 'r';
    end

    plot(t, data_clean, line_color, 'LineWidth', 1.2); hold on;

    title(title_str); ylabel('uV'); grid on;
    xlim([0 max(t)]); ylim(zoom_ylim);

    for i = 1:size(events,1)
        event_sec = (events{i,1} - current_interval_start) / fs;
        xline(event_sec, '--', 'Color', events{i,3}, 'LineWidth', 2);
    end
end

h = zeros(size(events,1), 1);
for i = 1:size(events,1)
    h(i) = plot(NaN, NaN, '--', 'Color', events{i,3}, 'LineWidth', 2);
end

lgd = legend(h, events(:,2), 'Orientation', 'horizontal', 'Location',
'southoutside');
lgd.FontSize = 10;
lgd.Box = 'on';
```

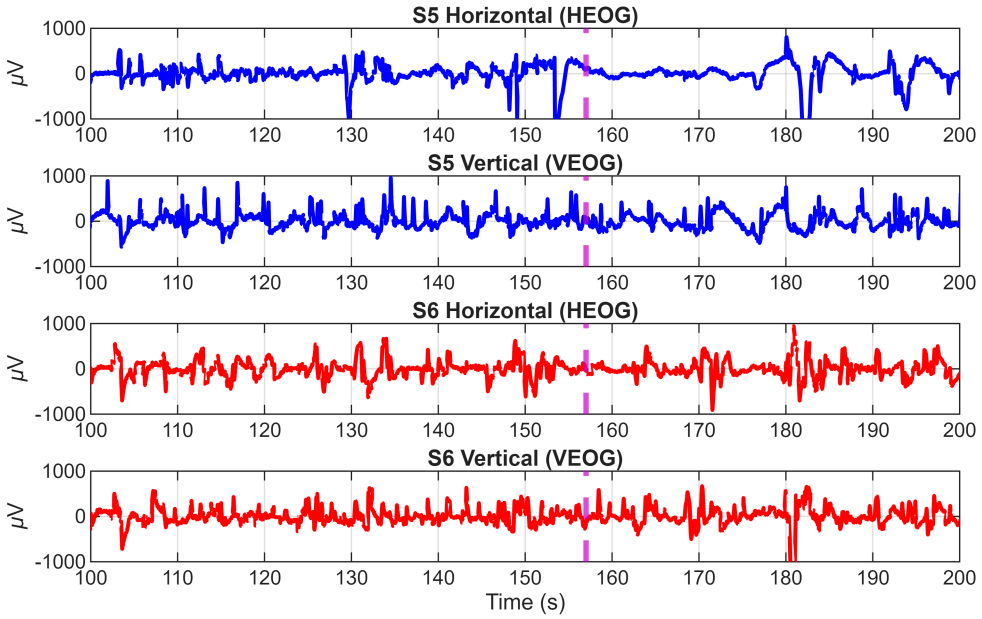


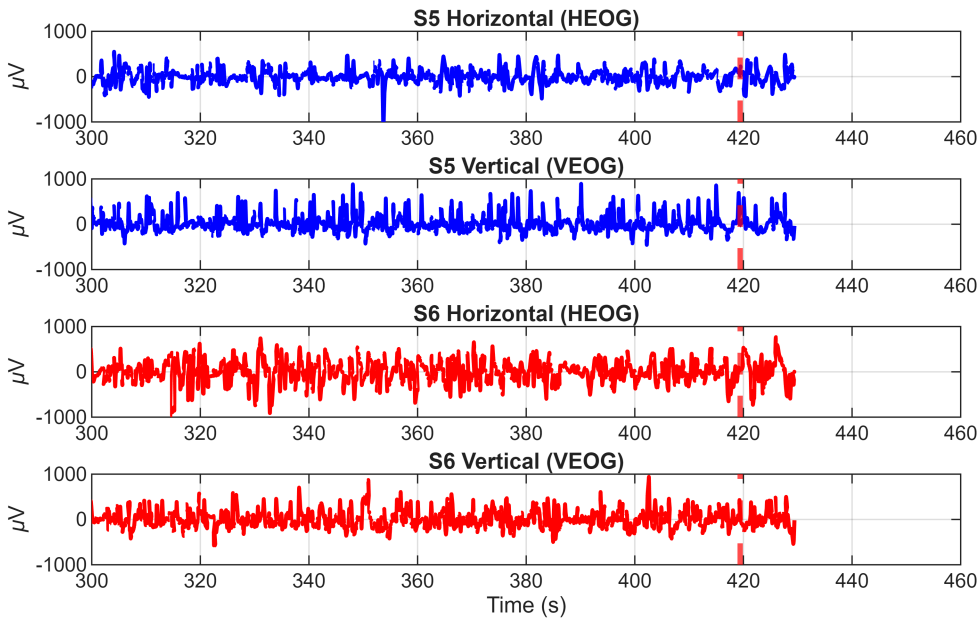
Zoom into marker zones to obtain 3 graphs

```
zoom_settings = {
    0,   80,  'Zoom 1: Dance Start (0-80s)';
    100, 200, 'Zoom 2: Transition (100-200s)';
    300, 460, 'Zoom 3: End Sequence (400-430s)'
```

```
};

for z = 1:size(zoom_settings, 1)
    z_start = zoom_settings{z, 1};
    z_end   = zoom_settings{z, 2};
    z_title = zoom_settings{z, 3};

    figure('Color','w', 'Name', z_title);

    for sp = 1:4
        subplot(4,1,sp);

        if sp == 1
            data = heog_s5_clean;
            t_str = 'S5 Horizontal (HEOG)';
            col = 'b';
        elseif sp == 2
            data = veog_s5_clean;
            t_str = 'S5 Vertical (VEOG)';
            col = 'b';
        elseif sp == 3
            data = heog_s6_clean;
            t_str = 'S6 Horizontal (HEOG)';
            col = 'r';
        elseif sp == 4
            data = veog_s6_clean;
            t_str = 'S6 Vertical (VEOG)';
            col = 'r';
        end

        plot(t, data, col, 'LineWidth', 1.5); hold on;
        ylabel('\muV');
        title(t_str);

        if sp == 4
            xlabel('Time (s)');
        end

        xlim([z_start z_end]);
        ylim([-1000 1000]);
        grid on;

        for i = 1:size(events,1)
            event_sec = (events{i,1} - current_interval_start) / fs;
            if event_sec >= z_start && event_sec <= z_end
                xline(event_sec, '--', 'Color', events{i,3}, 'LineWidth', 2);
            end
        end
    end
end
```


S5 Horizontal (HEOG)
S5 Vertical (VEOG)
S6 Horizontal (HEOG)
S6 Vertical (VEOG)
μV
1000
0
-1000
0
10
20
30
40
50
60
70
80
Time (s)


S5 Horizontal (HEOG)
S5 Vertical (VEOG)
S6 Horizontal (HEOG)
S6 Vertical (VEOG)
μV
1000
0
-1000
100
110
120
130
140
150
160
170
180
190
200
Time (s)



Z-score Thresholding to find blinks

1. Apply a bandpass filter from 0.5 to 10 Hz
2. Apply zero phase filtering
3. Convert voltage to standard deviation because we can't apply a single threshold to all
4. Catch absolute value because blinks can be positive or negative
5. 3 sigma rule because 99.7% of data points fall within ±3 standard deviations in a normal distribution
6. Apply a mask to then move on with saccades

```
subjects = {'S5', 'S6'};
metrics = struct();
fs = 1000;

for s = 1:2
    subj = subjects{s};

    if strcmp(subj, 'S5')
        h_data = heog_s5_clean(:);
        v_data = veog_s5_clean(:);
    else
        h_data = heog_s6_clean(:);
        v_data = veog_s6_clean(:);
    end

    % Filter specifically to find blink 0.5 - 10 Hz
    [b_blink, a_blink] = butter(2, [0.5 10]/(fs/2));
    v_blink_temp = filtfilt(b_blink, a_blink, double(v_data));

    % Normalize to Z-Score
```

```
v_z = (v_blink_temp - mean(v_blink_temp)) / std(v_blink_temp);

% Detect Peaks > 3 Std Devs
[~, locs] = findpeaks(abs(v_z), 'MinPeakHeight', 3, 'MinPeakDistance', 0.2*fs);

% Create Blink Mask
blink_mask = false(size(v_data));
win_pre = round(0.1 * fs);
win_post = round(0.15 * fs);

for k = 1:length(locs)
    p_start = max(1, locs(k) - win_pre);
    p_end   = min(length(v_data), locs(k) + win_post);
    blink_mask(p_start:p_end) = true;
end
```

I-VT Threshold to find Saccades

1. Calculate point-to-point velocities for each point in the protocol
2. Smooth
3. Label each point below velocity threshold as a fixation point, otherwise as a saccade point

It uses a FIR filter

```
vx = [0; diff(h_data)] * fs;
vy = [0; diff(v_data)] * fs;
vel_raw = sqrt(vx.^2 + vy.^2);

vel_smooth = smoothdata(vel_raw, 'gaussian', 20);

clean_velocity = vel_smooth(~blink_mask);
vt_thresh = mean(clean_velocity) + (5 * std(clean_velocity));

saccade_mask = (vel_smooth > vt_thresh) & (~blink_mask);
fixation_mask = (vel_smooth <= vt_thresh) & (~blink_mask);
```

Measure properties and metrics

```
% 1. Blinks
edges_b = diff([0; blink_mask; 0]);
metrics.(subj).Blinks.Starts = find(edges_b == 1);
metrics.(subj).Blinks.Ends   = find(edges_b == -1) - 1;

% 2. Saccades
edges_s = diff([0; saccade_mask; 0]);
metrics.(subj).Saccades.Starts = find(edges_s == 1);
metrics.(subj).Saccades.Ends   = find(edges_s == -1) - 1;
```

```
    metrics.(subj).Saccades.VelocityTrace = vel_smooth;

    % 3. Fixations
    edges_f = diff([0; fixation_mask; 0]);
    metrics.(subj).Fixations.Starts = find(edges_f == 1);
    metrics.(subj).Fixations.Ends   = find(edges_f == -1) - 1;

    % 4. Measure Saccade Properties
    num_sac = length(metrics.(subj).Saccades.Starts);
    metrics.(subj).Saccades.Amplitudes = zeros(num_sac, 1);
    metrics.(subj).Saccades.Durations  = zeros(num_sac, 1);
    metrics.(subj).Saccades.PeakVels   = zeros(num_sac, 1);

    for k = 1:num_sac
        idx = metrics.(subj).Saccades.Starts(k) : metrics.(subj).Saccades.Ends(k);

        dx = h_data(idx(end)) - h_data(idx(1));
        dy = v_data(idx(end)) - v_data(idx(1));

        metrics.(subj).Saccades.Amplitudes(k) = sqrt(dx^2 + dy^2);
        metrics.(subj).Saccades.Durations(k)  = length(idx) / fs;
        metrics.(subj).Saccades.PeakVels(k)   = max(vel_smooth(idx));
    end

    % Measure Fixations
    num_fix = length(metrics.(subj).Fixations.Starts);
    metrics.(subj).Fixations.Durations = zeros(num_fix, 1);
    for k = 1:num_fix
        idx = metrics.(subj).Fixations.Starts(k) : metrics.(subj).Fixations.Ends(k);
        metrics.(subj).Fixations.Durations(k) = length(idx) / fs;
    end

    fprintf('Subject %s (Hybrid): Found %d Blinks (Z-Score) and %d Saccades (I-
VT).\n', ...
        subj, length(locs), num_sac);
end
```

```
Subject S5 (Hybrid): Found 183 Blinks (Z-Score) and 75 Saccades (I-VT).
Subject S6 (Hybrid): Found 85 Blinks (Z-Score) and 140 Saccades (I-VT).
```

Box plot and raw stats for metric comparision

```
% Function to get rate (Hz)
get_rate = @(starts, total_dur) length(starts) / total_dur;

% Total duration of the recording
total_time_s5 = length(metrics.S5.Saccades.VelocityTrace) / fs;
total_time_s6 = length(metrics.S6.Saccades.VelocityTrace) / fs;
```

SACCADES

```
count_s5 = length(metrics.S5.Saccades.Starts);
count_s6 = length(metrics.S6.Saccades.Starts);
rate_s5  = count_s5 / total_time_s5;
rate_s6  = count_s6 / total_time_s6;

fprintf('SACCADES:\n');
```

```
SACCADES:
```

```
fprintf('  S5 Count: %d | Rate: %.2f Hz\n', count_s5, rate_s5);
```

```
  S5 Count: 75 | Rate: 0.17 Hz
```

```
fprintf('  S6 Count: %d | Rate: %.2f Hz\n', count_s6, rate_s6);
```

```
  S6 Count: 140 | Rate: 0.33 Hz
```

```
fprintf('----------------------------------------\n');
```

```
--------------------------------------
```

FIXATIONS

```
count_f5 = length(metrics.S5.Fixations.Starts);
count_f6 = length(metrics.S6.Fixations.Starts);

fprintf('FIXATIONS:\n');
```

```
FIXATIONS:
```

```
fprintf('  S5 Count: %d\n', count_f5);
```

```
  S5 Count: 257
```

```
fprintf('  S6 Count: %d\n', count_f6);
```

```
  S6 Count: 219
```

```
fprintf('----------------------------------------\n');
```

```
--------------------------------------
```

BLINKS

```
count_b5 = length(metrics.S5.Blinks.Starts);
count_b6 = length(metrics.S6.Blinks.Starts);
rate_b5  = count_b5 / total_time_s5 * 60;
rate_b6  = count_b6 / total_time_s6 * 60;

fprintf('BLINKS:\n');
```

```
BLINKS:
```

```
fprintf('  S5 Count: %d | Rate: %.2f blinks/min\n', count_b5, rate_b5);
```

```
  S5 Count: 183 | Rate: 25.57 blinks/min
```

```
fprintf('  S6 Count: %d | Rate: %.2f blinks/min\n', count_b6, rate_b6);
```

```
  S6 Count: 82 | Rate: 11.46 blinks/min
```

```
fprintf('========================================\n\n');
```

```
========================================
```

Generate distribution plots

```
figure('Color', 'w', 'Name', 'Metric Distributions: S5 vs S6', 'Position', [100 100
1200 600]);
t = tiledlayout(2, 3, 'TileSpacing', 'compact', 'Padding', 'compact');

c5 = [0.2 0.6 1];
c6 = [1 0.4 0.4];
```

Obtain metrics such as amplitude, duration and peak velocity

```
data_s5 = metrics.S5.Saccades.Amplitudes;
data_s6 = metrics.S6.Saccades.Amplitudes;
create_boxplot(nexttile, data_s5, data_s6, 'Saccade Amplitude', '\muV');
```

```
Warning: boxplot might not be displayed properly in the tiled chart layout.
```

```
data_s5 = metrics.S5.Saccades.Durations * 1000; % ms
data_s6 = metrics.S6.Saccades.Durations * 1000;
create_boxplot(nexttile, data_s5, data_s6, 'Saccade Duration', 'ms');
```

```
Warning: boxplot might not be displayed properly in the tiled chart layout.
```

```
data_s5 = metrics.S5.Saccades.PeakVels;
data_s6 = metrics.S6.Saccades.PeakVels;
create_boxplot(nexttile, data_s5, data_s6, 'Peak Velocity', '\muV/s');
```

```
Warning: boxplot might not be displayed properly in the tiled chart layout.
```

Additional metrics such as fixation duration, blink duration

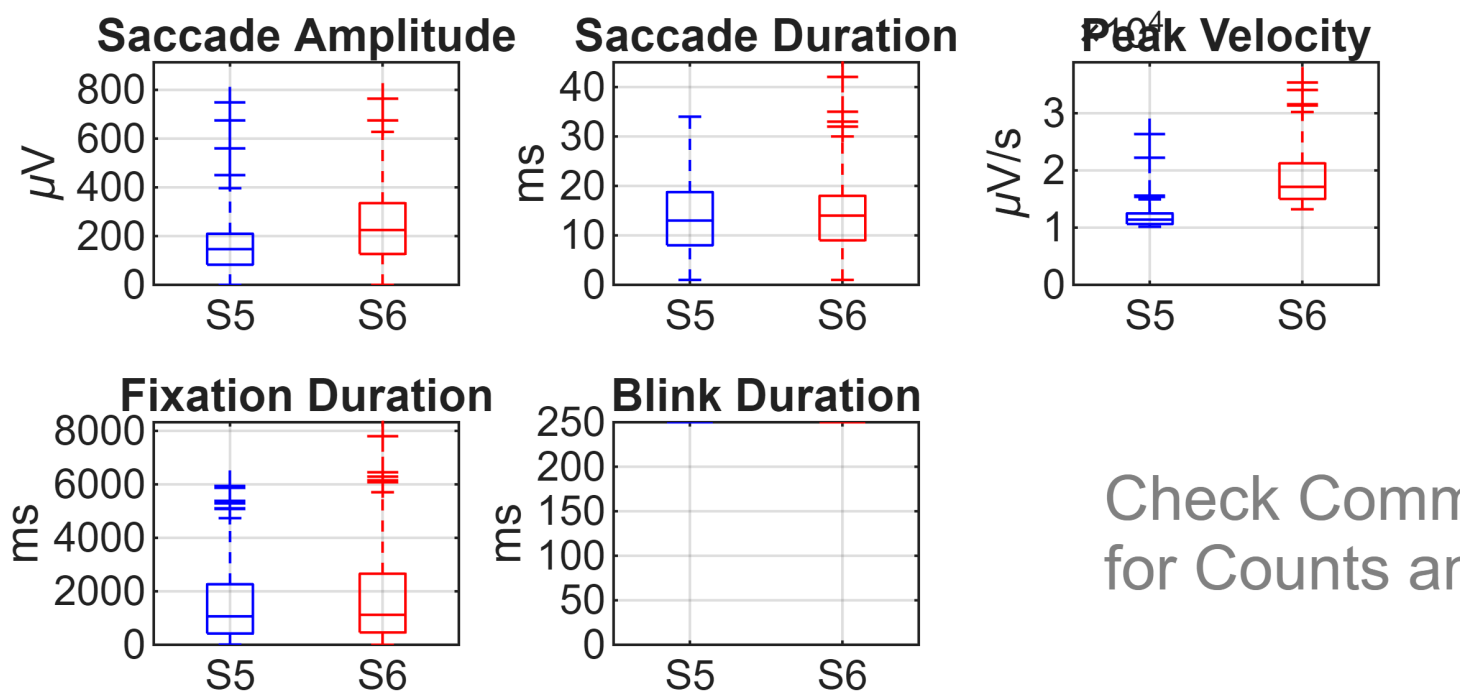
```
data_s5 = metrics.S5.Fixations.Durations * 1000;
data_s6 = metrics.S6.Fixations.Durations * 1000;
create_boxplot(nexttile, data_s5, data_s6, 'Fixation Duration', 'ms');
```

```
Warning: boxplot might not be displayed properly in the tiled chart layout.
```

```
dur_b5 = (metrics.S5.Blinks.Ends - metrics.S5.Blinks.Starts) / fs * 1000;
dur_b6 = (metrics.S6.Blinks.Ends - metrics.S6.Blinks.Starts) / fs * 1000;
create_boxplot(nexttile, dur_b5, dur_b6, 'Blink Duration', 'ms');
```

```
Warning: boxplot might not be displayed properly in the tiled chart layout.
```

```
nexttile; axis off;
text(0.1, 0.5, {'Check Command Window', 'for Counts and Rates'}, 'FontSize', 12,
'Color', [0.5 0.5 0.5]);
```



Helper function for box plot

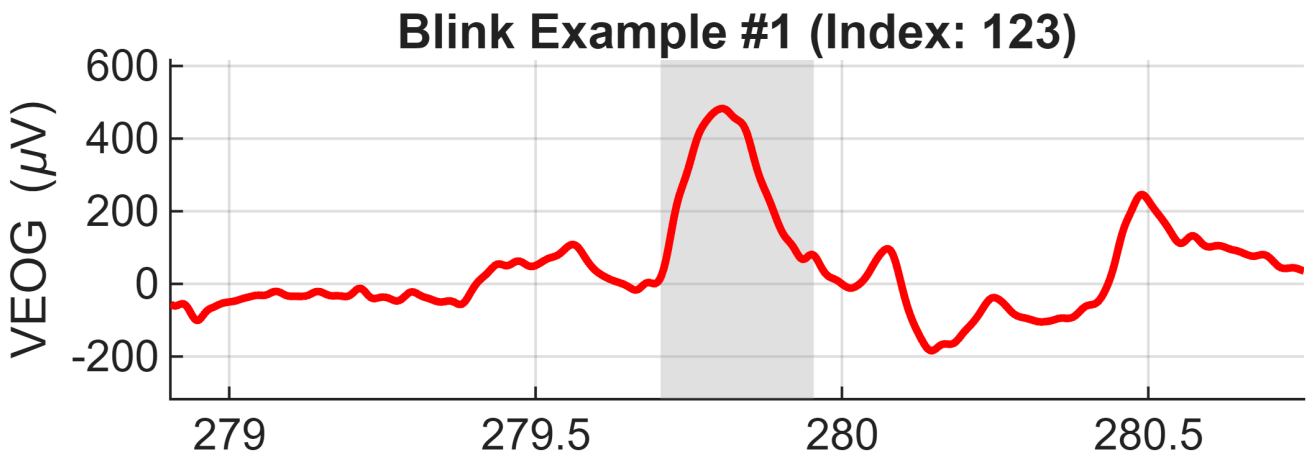
```
function create_boxplot(ax, d5, d6, title_str, unit_str)
    if isempty(d5), d5 = NaN; end
    if isempty(d6), d6 = NaN; end

    group_ids = [repmat({'S5'}, length(d5), 1); repmat({'S6'}, length(d6), 1)];
    data_all = [d5; d6];

    boxplot(ax, data_all, group_ids, 'Colors', 'br', 'Symbol', '+');
    title(title_str);
    ylabel(unit_str);
    grid on;
    q3 = quantile(data_all, 0.75);
    iqr_val = iqr(data_all);
    upper_lim = q3 + 3 * iqr_val;
```

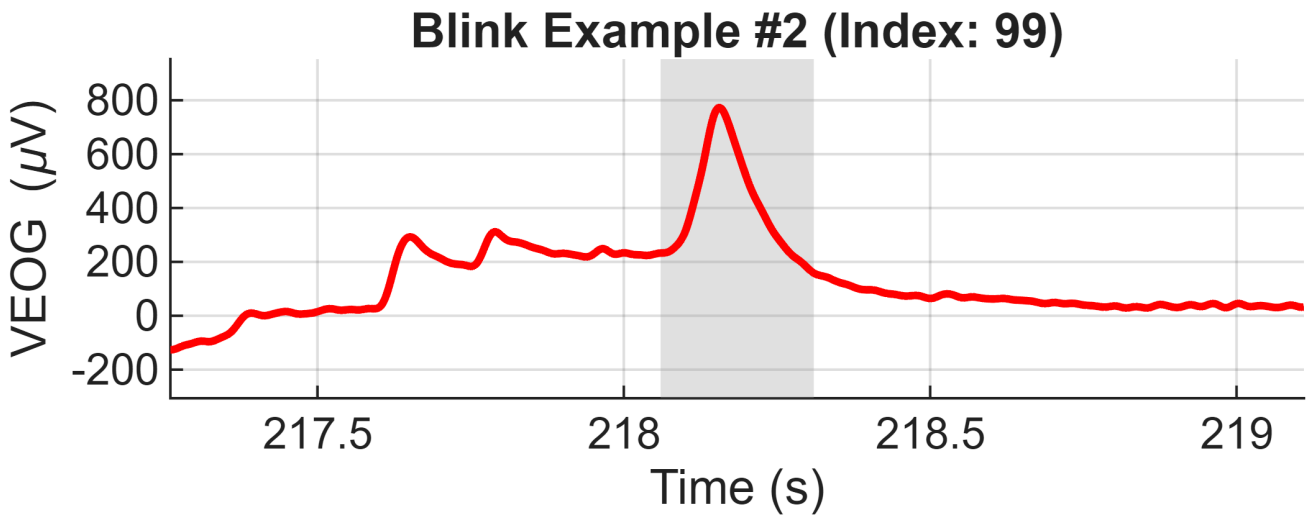
```
    if ~isnan(upper_lim) && max(data_all) > upper_lim
        ylim([min(0, min(data_all)) upper_lim]);
    end
end
```

Plot elements and highlight them in gray

Two random Blinks

```
subj = 'S5';
if ~isfield(metrics, subj)
    error('Metrics for %s not found. Run the extraction step first.', subj);
end

h_sig = heog_s5_clean(:);
v_sig = veog_s5_clean(:);
t_full = (0:length(h_sig)-1) / fs;

pad_sec = 0.8;
pad_samp = round(pad_sec * fs);

evts = metrics.(subj).Blinks;
indices = find_random_valid_indices(evts, 2, length(h_sig), pad_samp);

figure('Color', 'w', 'Name', 'Random Blink Examples', 'Position', [50 100 800 600]);
tiledlayout(2, 1, 'TileSpacing', 'loose');

for k = 1:2
    idx = indices(k);
    w_start = evts.Starts(idx) - pad_samp;
    w_end   = evts.Ends(idx) + pad_samp;
    range   = w_start:w_end;

    nexttile; hold on;
    plot(t_full(range), v_sig(range), 'r', 'LineWidth', 1.5);

    e_start = t_full(evts.Starts(idx));
    e_end   = t_full(evts.Ends(idx));
    xregion(e_start, e_end, 'FaceColor', [0.6 0.6 0.6], 'FaceAlpha', 0.3);

    title(sprintf('Blink Example #%d (Index: %d)', k, idx));
    ylabel('VEOG (\muV)'); grid on;

    axis tight;
    yl = ylim;
    margin = (yl(2) - yl(1)) * 0.2;
    ylim([yl(1)-margin yl(2)+margin]);

    if k==2, xlabel('Time (s)'); end
end
```

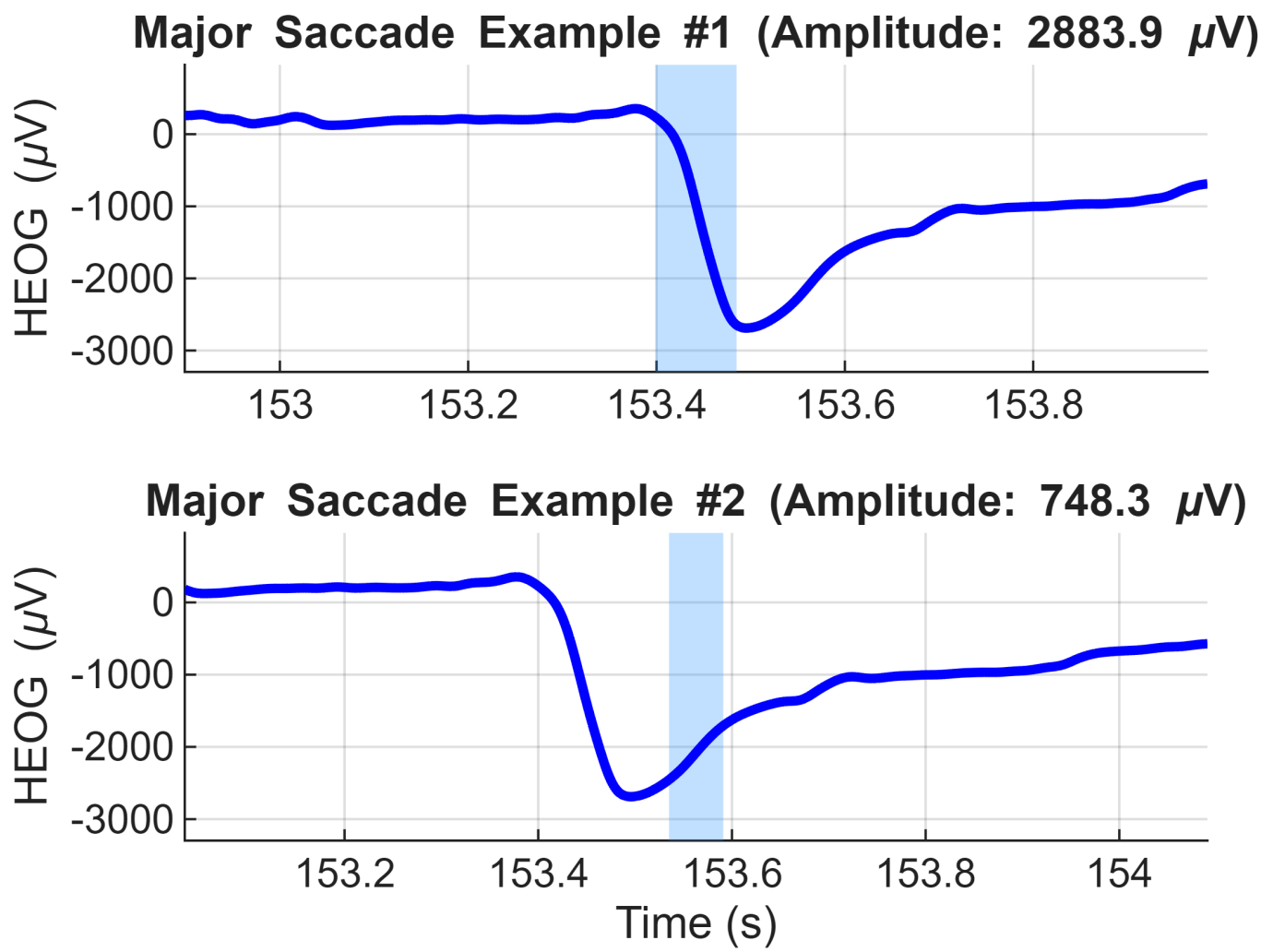




Two large saccades

```
subj = 'S5';
h_sig = heog_s5_clean(:);
t_full = (0:length(h_sig)-1) / fs;
evts = metrics.(subj).Saccades;

[sorted_amps, sort_idx] = sort(evts.Amplitudes, 'descend');

top_indices = [];
pad_sec = 0.5;
pad_samp = round(pad_sec * fs);

for k = 1:length(sort_idx)
    orig_idx = sort_idx(k);
    s = evts.Starts(orig_idx);
    e = evts.Ends(orig_idx);
```

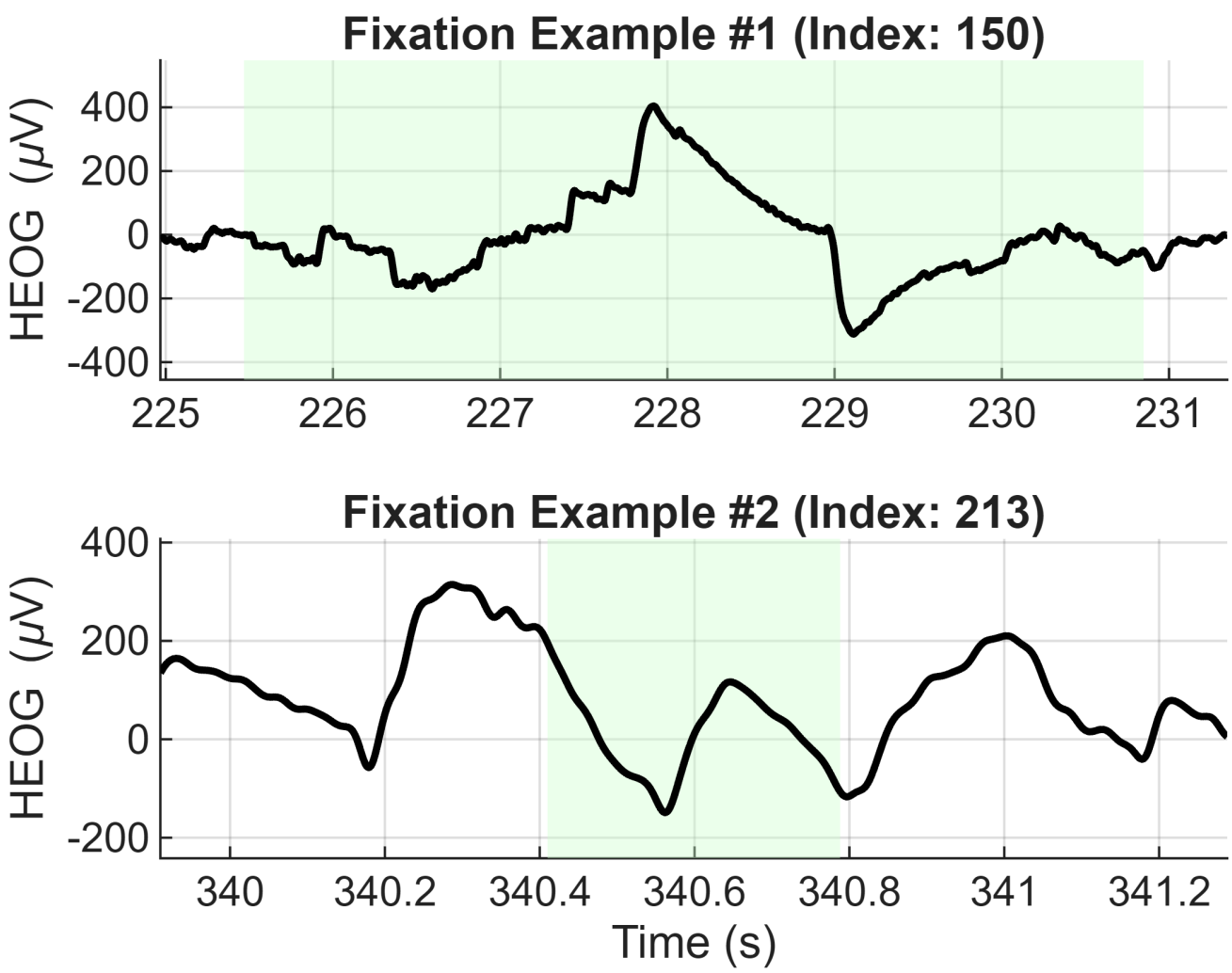
```
    if (s - pad_samp > 1) && (e + pad_samp < length(h_sig))
        top_indices(end+1) = orig_idx; %#ok<AGROW>
    end
    if length(top_indices) >= 2, break; end
end

figure('Color', 'w', 'Name', 'Textbook Saccade Examples', 'Position', [100 100 800
600]);
tiledlayout(2, 1, 'TileSpacing', 'loose');

for k = 1:length(top_indices)
    idx = top_indices(k);

    w_start = evts.Starts(idx) - pad_samp;
    w_end   = evts.Ends(idx) + pad_samp;
    range   = w_start:w_end;

    nexttile; hold on;
    plot(t_full(range), h_sig(range), 'b', 'LineWidth', 2);

    e_start = t_full(evts.Starts(idx));
    e_end   = t_full(evts.Ends(idx));
    xregion(e_start, e_end, 'FaceColor', [0.2 0.6 1], 'FaceAlpha', 0.3);

    amp = evts.Amplitudes(idx);
    title(sprintf('Major Saccade Example #%d (Amplitude: %.1f \\muV)', k, amp));
    ylabel('HEOG (\muV)'); grid on; axis tight;
    yl = ylim; margin = (yl(2)-yl(1))*0.2;
    ylim([yl(1)-margin yl(2)+margin]);

    if k==2, xlabel('Time (s)'); end
end
```



Two random fixations

```
evts = metrics.(subj).Fixations;
indices = find_random_valid_indices(evts, 2, length(h_sig), pad_samp);

figure('Color', 'w', 'Name', 'Random Fixation Examples', 'Position', [150 100 800
600]);
tiledlayout(2, 1, 'TileSpacing', 'loose');

for k = 1:2
    idx = indices(k);
    w_start = evts.Starts(idx) - pad_samp;
    w_end   = evts.Ends(idx) + pad_samp;
    range   = w_start:w_end;

    nexttile; hold on;
    plot(t_full(range), h_sig(range), 'k', 'LineWidth', 1.5);

    e_start = t_full(evts.Starts(idx));
```

```
    e_end   = t_full(evts.Ends(idx));
    xregion(e_start, e_end, 'FaceColor', [0.8 1 0.8], 'FaceAlpha', 0.4);

    title(sprintf('Fixation Example #%d (Index: %d)', k, idx));
    ylabel('HEOG (\muV)'); grid on;

    axis tight;
    yl = ylim;
    margin = (yl(2) - yl(1)) * 0.2;
    ylim([yl(1)-margin yl(2)+margin]);

    if k==2, xlabel('Time (s)'); end
end
```



Helper random function selection

```
function indices = find_random_valid_indices(evt_struct, count, data_len, padding)
    indices = [];
    found = 0;
```

```
    num_events = length(evt_struct.Starts);
    if num_events == 0
        warning('No events found for this category.');
        return;
    end

    random_order = randperm(num_events);

    for k = 1:num_events
        idx = random_order(k);

        s = evt_struct.Starts(idx);
        e = evt_struct.Ends(idx);

        if (s - padding > 1) && (e + padding < data_len)
            indices = [indices, idx]; %#ok<AGROW>
            found = found + 1;
        end

        if found >= count
            break;
        end
    end

    if found < count
        warning('Could not find enough valid events. Using whatever was found.');
    end
end
```

Synchrony visualization

```
v5 = metrics.S5.Saccades.VelocityTrace;
v6 = metrics.S6.Saccades.VelocityTrace;

min_len = min(length(v5), length(v6));
v5 = v5(1:min_len);
v6 = v6(1:min_len);
t = (1:min_len) / fs;

figure('Color', 'w', 'Name', 'Comprehensive Synchrony Analysis', 'Position', [50 50
1200 900]);
tiledlayout(2, 2, 'TileSpacing', 'compact', 'Padding', 'compact');
```

PLOT B: CROSS-CORRELATION VS LAG

Interpretation: A peak at 0 means perfect sync. A shift means one reacts to the other.

```
nexttile;
```

```
max_lag_sec = 2.0;
max_lag = round(max_lag_sec * fs);
[r, lags] = xcorr(v5 - mean(v5), v6 - mean(v6), max_lag, 'coeff');
lags_ms = (lags / fs) * 1000;

plot(lags_ms, r, 'k', 'LineWidth', 1.5); hold on;

[max_r, idx] = max(r);
peak_lag = lags_ms(idx);
xline(peak_lag, 'r--', sprintf('Peak: %.0f ms', peak_lag),
'LabelVerticalAlignment', 'bottom');
xline(0, 'k-', 'Alpha', 0.3);

title('B. Cross-Correlation Function');
xlabel('Lag (ms) [Negative: S6 Leads | Positive: S5 Leads]');
ylabel('Correlation Coefficient (r)');
grid on; axis tight;
```

Scatter plot for velocity

A diagonal "cigar" shape = Synchronized. A round blob = Independent.

```
nexttile;

ds = 10;
scatter(v5(1:ds:end), v6(1:ds:end), 10, 'filled', 'MarkerFaceColor', [0.2 0.2 0.6],
'MarkerFaceAlpha', 0.1);
hold on;

max_val = max([max(v5), max(v6)]);
plot([0 max_val], [0 max_val], 'r--', 'LineWidth', 1.5);

title('C. Instantaneous Velocity Coupling');
xlabel('S5 Velocity (\muV/s)');
ylabel('S6 Velocity (\muV/s)');
axis square; grid on;
xlim([0 max_val*0.8]); ylim([0 max_val*0.8]);
```

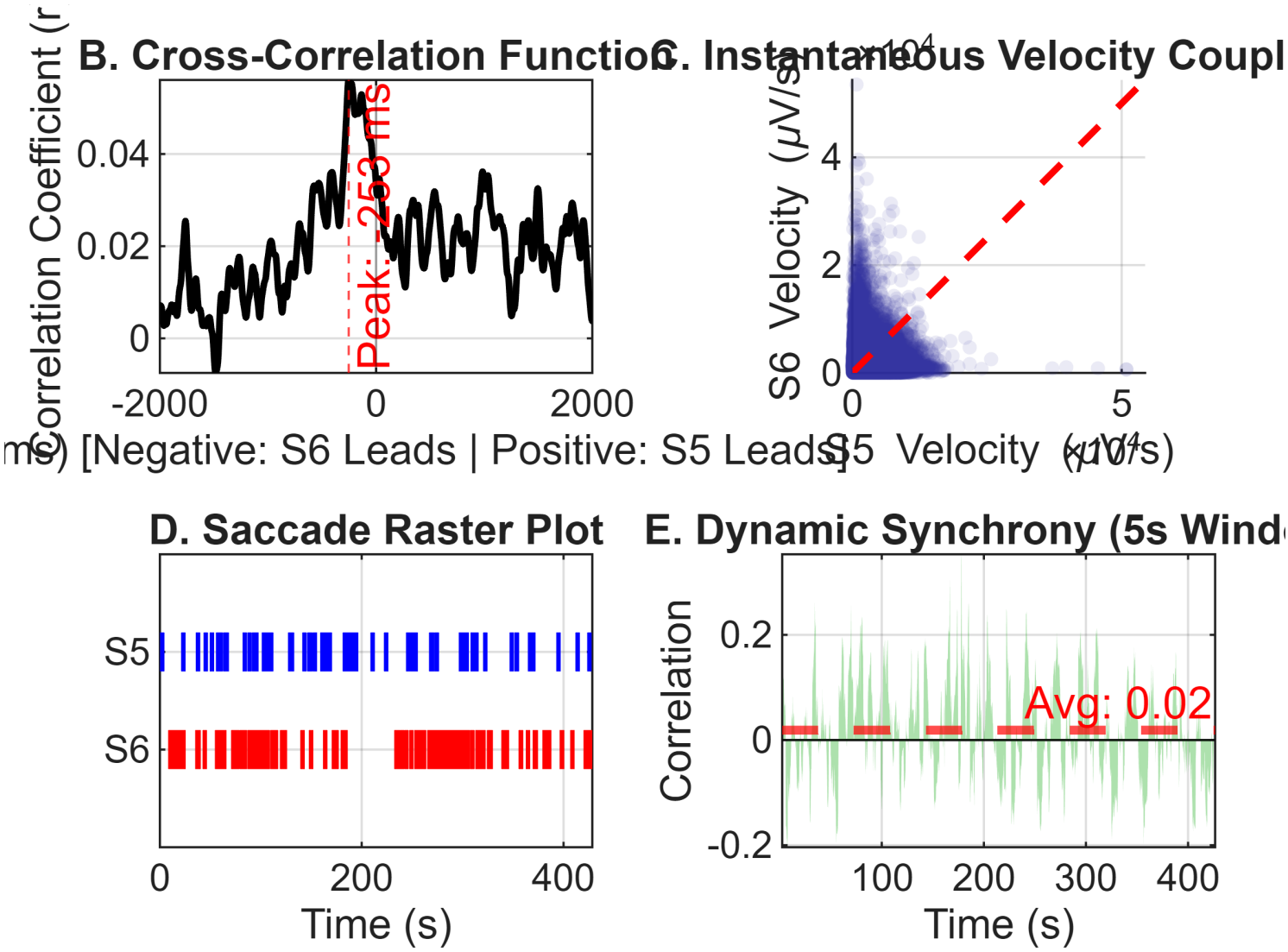
Aligned saccade raster plot

```
nexttile;

starts_s5 = metrics.S5.Saccades.Starts / fs;
starts_s6 = metrics.S6.Saccades.Starts / fs;

for i = 1:length(starts_s5)
    plot([starts_s5(i) starts_s5(i)], [1.8 2.2], 'b', 'LineWidth', 1); hold on;
end
```

```
for i = 1:length(starts_s6)
    plot([starts_s6(i) starts_s6(i)], [0.8 1.2], 'r', 'LineWidth', 1);
end

title('D. Saccade Raster Plot');
xlabel('Time (s)');
ylim([0 3]);
yticks([1 2]); yticklabels({'S6', 'S5'});
grid on;
```

Correlation

1. Apply sliding window approach of 5 seconds
2. Do step overlapping, by stepping forward only 0.25 seconds at a time (overlapping the previous window by 4.75 seconds
3. Calculate the correlation metric with pearson coefficient R
4. Handling 0 variance edge case, if one person is moving and the other one isn't we get R=0
5. Display in a time series plot with R as the area of the graph

```
nexttile;

win_sec = 5;
win_len = round(win_sec * fs);
step_size = round(fs / 4);

num_steps = floor((length(v5) - win_len) / step_size);
r_windowed = zeros(num_steps, 1);
t_windowed = zeros(num_steps, 1);

for i = 1:num_steps
    idx_start = (i-1)*step_size + 1;
    idx_end = idx_start + win_len - 1;

    chunk5 = v5(idx_start:idx_end);
    chunk6 = v6(idx_start:idx_end);

    R = corrcoef(chunk5, chunk6);

    if isnan(R(1,2))
        r_windowed(i) = 0;
    else
        r_windowed(i) = R(1,2);
    end

    t_windowed(i) = t(round((idx_start + idx_end)/2));
end

% 3. Plot The Result
```

```
area(t_windowed, r_windowed, 'FaceColor', [0.2 0.7 0.2], 'FaceAlpha', 0.4,
'EdgeColor', 'none');
hold on;
yline(0, 'k-');
mean_sync = mean(r_windowed);
yline(mean_sync, 'r--', sprintf('Avg: %.2f', mean_sync), 'LineWidth', 2);

title('E. Dynamic Synchrony (5s Window)');
xlabel('Time (s)');
ylabel('Correlation');
ylim([-0.5 1]);
grid on; axis tight;
```